

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **back-end AI-powered applications** using **Python, Flask, and FastAPI**, integrating **LLM orchestration frameworks** including **LangChain** and **Semantic Kernel** for multi-agent system automation; achieved **40% reduction** in manual workflow processing time.
- Built and deployed **REST and GraphQL APIs** for AI backend services on **AWS (Lambda, SageMaker)** and **Azure ML**, enabling seamless front-end integration with **React, Angular, and Vue.js** interfaces; managed **5+ concurrent production AI services** with **99.5% uptime**.
- Implemented **MLOps practices** for model deployment, monitoring, and lifecycle management using **TensorFlow** and **PyTorch**; established automated **unit, integration, and end-to-end testing** frameworks with **pytest** and **Jest**, reducing deployment defects by **35%**.
- Executed **prompt engineering** and experimentation across multiple AI models using **Claude API, GPT-4, and Azure OpenAI**; documented model strengths, limitations, and data-driven justifications; applied **constraint guardrails** and **structured JSON** responses for production reliability.
- Designed and deployed secure, compliant AI systems adhering to **GDPR, EU AI Act, and Responsible AI guidelines**; conducted security audits for LLM integrations and applied **data protection best practices** across all AI application tiers.
- Mentored junior developers through pair programming, code reviews, and knowledge-sharing sessions on **AI architecture, API integration patterns, and Agile/Scrum practices**; authored comprehensive technical documentation on AI model capabilities, regulatory compliance, and system design decisions.

SKILLS

Backend AI & LLM Development – Python, Flask, FastAPI, LangChain, Semantic Kernel, multi-agent systems orchestration, Claude API, GPT-4, Azure OpenAI.

Machine Learning & ML Frameworks – TensorFlow, PyTorch, scikit-learn, pandas, NumPy, model training, evaluation metrics, supervised learning, unsupervised learning.

Cloud ML & Deployment – AWS SageMaker, Azure ML, AWS Lambda, EC2, S3, CloudWatch, model lifecycle management, MLOps, automated deployment pipelines.

Frontend Development & Integration – React, Angular, Vue.js, HTML, CSS, JavaScript, TypeScript, REST APIs, GraphQL, Figma, Storybook, accessible UI design.

Testing & Quality Assurance – pytest, Jest, unit testing, integration testing, end-to-end testing, automated test frameworks, test-driven development.

Prompt Engineering & AI Security – prompt engineering, constraint guardrails, structured-JSON prompting, GDPR compliance, EU AI Act, Responsible AI practices, AI security best practices.

Development & Agile Practices – Agile methodologies, Scrum, Application Lifecycle Management, Git, Jenkins CI/CD, Jira, code reviews, documentation, mentoring.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

PROJECTS

Ireland Jobs Tracker – AI Resume Generation Engine (Personal Project)

- Engineered end-to-end **LLM orchestration workflow** integrating **Claude API** (claude-haiku-4-5) with **Apify web scrapers** for automated job description ingestion and processing.
- Developed **prompt engineering framework** with **structured-JSON constraint guardrails** to generate JD-tailored resume content at scale; optimized token efficiency and output consistency across **100+ generated resumes**.
- Built complete automation pipeline: JD ingestion → **Claude API** calls → **reportlab PDF generation** → **GitHub Actions CI/CD** → **Vercel deployment**; achieved fully automated resume generation workflow with **zero manual intervention**.
- Applied **prompt engineering best practices**, tested multiple model parameters, and implemented guardrails for output quality; documented model limitations and validated resume accuracy against source job descriptions.

SWIFT Payments & Multi-Agent Dashboard – AI-Enhanced Operations (HCL Technologies)

- Designed and deployed **backend AI services** for SWIFT messaging payment operations using **Python**, **FastAPI**, and **LangChain**-based multi-agent orchestration to automate payment-flow analysis and anomaly detection.
- Integrated **LLM models** (Claude, GPT-4) for intelligent transaction classification, regulatory compliance assessment, and automated SLA breach prediction; achieved **45% reduction** in manual payment reconciliation time.
- Built **React-based front-end interfaces** integrated with **REST API** backend services for real-time payment dashboard visualization, **ISO 20022 message flow** transparency, and operational alerting; deployed on **AWS Lambda** and **Azure ML**.
- Implemented **GDPR** and **EU AI Act** compliance measures across LLM workflows; applied **Responsible AI practices** for model bias detection and financial data protection; maintained comprehensive documentation of AI model strengths, limitations, and regulatory justifications.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)